

PEST AND DISEASES OF CROPS AND LIVESTOCK

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DEFINITION

A pest is an organism which harbors disease organism(s) or causes damage to other organism(s). There are crop and livestock (animals) pests. Crop pests include insects such as grasshoppers, mealy bugs, myriads, beetles, birds and mammals (such as rodents) while livestock pests are ectoparasites such as ticks, mites and endoparasites such as liver flukes, round worms and tapeworms. They can also be plant pests known as weeds or animal pests such as insects, birds, rodents, monkeys, man or nematodes.

TYPES OF PEST

Insects pest: These are arthropods that carry diseases or cause damage to plant and animals. Examples include: tick, lice, grasshoppers, cotton stainer etc.

Non insect pest: These are vertebrates and molluscs that common cause disease to plants and animals or destroy crops. E.g rodents, worms, nematodes, monkeys etc.

GROUPS OF INSECT CROP PESTS

Stem Borers: Stem borers of cereal crops like maize are the larvae of certain moths. They lay eggs at the junctions of leaf sheaths and stem of a maize plant which hatches after a week and weakens the stem causing the maize plant to break even in slight wind.

Fruit and Seed Feeders: These include fruit-piercing moths, fruit flies, cotton strainers and certain beetles. Examples are red boil-worm and the cotton stainer. Their host plant includes tomato, millet, maize and okra.

Root Feeders: They may be insect larvae or adults. An example is the yam beetle which lives mainly in the soil. They burrow into the ground and feed on yam tubers.

Leaf Feeders: Leaves of crops are eaten by snails, leaf-feeding beetles, caterpillars of various moths and butterflies, grasshoppers and locusts.

Young Shoot Feeders: Insects like aphids and mealy bug pierce and suck juices from young shoots of crop plants. They first settle on the apex and feed on the young tissue. Many sucking pests also transmit disease-causing fungi, bacteria, and virus to the plants e.g. cassava mosaic transmitted by white flies.

Life Cycle of Some Selected Pest Grasshopper

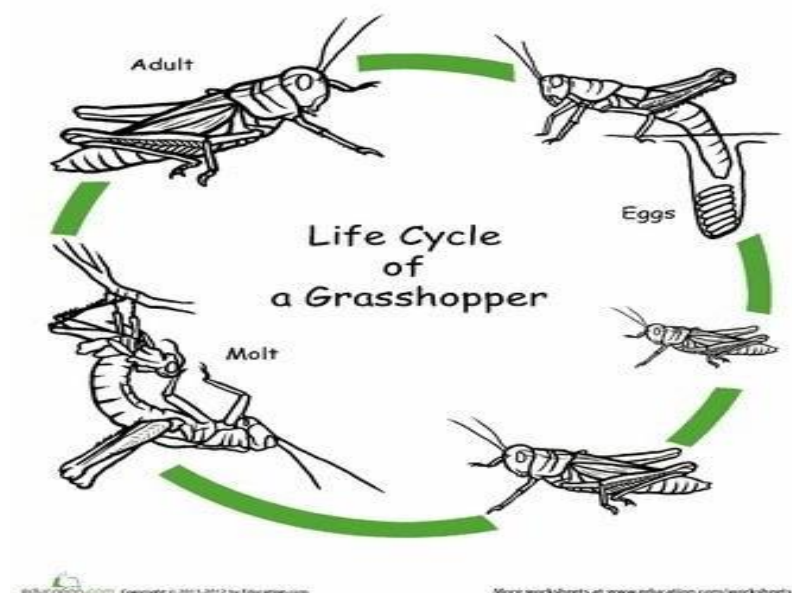
(Zonocerus variegates)

History: Male and female mate and eggs are then fertilized internally. The female then digs a hole with her ovipositor in the soil usually near the roots of plants and lays about 40 to 100 eggs the hole. A protective material is produced to cover the eggs. After about 3 days, the eggs hatch into nymphs. A nymph resembles the adult in all respect except it is sexually immature and wingless.

The nymph moults several times and hops about feeding on the shoots of plants

nearby. They begin to migrate into areas where there are cassava farms and start feeding on the cassava leaves until an adult stage is attained.

LIFE CYCLE OF A GRASSHOPPE



Effects of Grasshoppers on Cassava

Nymphs and adults feed on the leaves, shoots and barks of cassava thereby reducing cassava yield drastically as photosynthesis is seriously impaired.

Control: Spray with gammalin 20.

Cassava mealy bug (*Phenacoccus manihoti*)

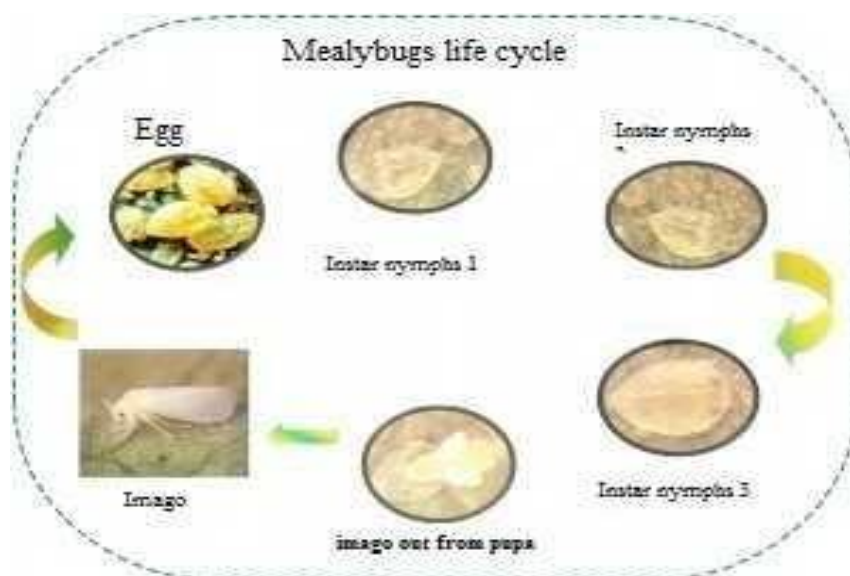
The female insect lays eggs without fertilization by the male. This type of reproduction is called parthenogenesis. The unfertilized eggs hatch into larvae which are wind borne or carried with cassava stem cuttings during planting because they hide in the buds of the stem. The larvae undergo three moulting stages before adult stage. One generation is completed in about twenty-two days and the adult has life span of about one hundred and forty-five days.

Effects

1. They suck the sap of cassava leading to the stunted growth of shoot.
2. The shoots develop bunchy tops.
3. The leaves of the shoot die and drop.
4. Mealybug infestation impairs photosynthesis

Control: Dip cassava cutting in 0.1% ultracide solution for one minute before planting tuber

The Life Cycle of Cassava Mealy bug



Life History of Bean Weevil

Male and female adults mate and fertilized eggs are then laid in the ripening pods in the farm. The eggs hatch into larvae which enter into the bean seeds before harvesting. In the store, the larvae feed on the cotyledons of the beans. The larvae develop into pupae in the store. The pupae later develop into adults which fly away to mate again in the farm.

Effects

The larvae feed on the seed thereby reducing the quality and value of the beans.

Control

Fumigate the store in which beans are stored with fumigants such as methyl bromide.

Cocoa Myriads (Capsid)

These are piercing and sucking insects. They attach the young shoot of cocoa, introducing toxic saliva into the sap which may kill the plant. It can also introduce viral diseases into the plant.

Effects

Cocoa myriads cause die back disease which reduces the growth of cocoa plant. Fruit yield is reduced.

Control

Spray the cocoa farm with kokotine or gammalin 20

Yam Beetle (*Heteroligusmeles*) Life

History

The mating of the female and the male yam beetles takes place between November and December in riverine areas and fertilized eggs are laid. Between December and February, eggs hatch into larvae that feed on the decayed organic substances. The larvae molt thrice before developing into pupae. In March, the pupae develop into adults after moulting. The adults then fly to areas where yams are planted between April and June. They dig into the soil to search for yam tubers. When they eventually come in contact with tubers, they feed on them and seriously damage the tubers. Between October and November, the adult yam beetles undergo breeding migration to the riverine areas again for mating.

Effects

Adult beetles feed on yam tubers causing serious damage to the tubers and render the tubers valueless. If the tubers are attacked at early stage, the yield becomes poor.

Control

Dust yam or yam sets with Aldrin before planting.

Diseases

A disease is a change of the plant or animal from the normal state of health, presenting marked symptom or outward visible signs in the organism. Disease pathogens are disease causing organisms, which include bacteria, viruses, fungi, nematodes and protozoan.

Diseases of Crops, Effects and Control

NAME OF DISEASE	CAUSATIVE ORGANISMS	SYMPTOMS	CONTROL
1. Maize rust	Fungus	Cob and grains covered with black spores, premature death.	Apply fungicides before planting.
2. Groundnut rosette	Virus	Leaves curl and turn yellow, show mottling and wrinkles, leads to death.	Uproot and burn infected plants. Plant resistant varieties
3. Rice blasts	Fungus	Brown spots on leaves, leaves dry up.	Use resistant variety, apply fungicide.
4. Cassava mosaic	Virus	Distorted leaves, yellow patches on leaves, stunted growth	Plant resistant varieties
5. Cocoa black pod	Fungus	Brown spots on pods. The pods	Apply fungicides.

	(Phytophytorapalmivora	turn black and dry up	Remove and burn all infected pods
6. Cassava wilt	Bacterium	Leaves wilt and fall off	Early planting, crop rotation
7. Cocoa swollen shoot	Virus	Stems and roots become swollen, leave chlorolysis	Burn infected plants, use resistant varieties

Evaluation

- Discuss one disease of cassava, cocoa and maize under the following sub headings
 - causative organism
 - mode of transmission
 - symptoms
 - control
- Mention the groups of disease pathogens.

Livestock Diseases, Effect and Control

NAME OF DISEASE	CAUSATIVE ORGANISMS	SYMPTOMS	CONTROL
1. New castle disease (poultry)	Virus	Neck twisting, paralysis of legs, respiratory and nervous disorder, low productivity and death	Sanitation, vaccination
2. Rinder pest (cattle)	Virus	High fever and diarrhea	Isolation of infected animal, vaccination
3. Anthrax (cattle, goat, sheep)	Bacteria	Swellings in the neck, and lower abdomen, fever, blood stained discharge from the nostrils	Immunizatio, sanitation
4. Coccidiosis (poultry)	Protozoan	Blood stool, affects intestines	Sanitation

5. Trypanosomiasis (cattle)	Protozoan (trypanosome)	Intermittent fever, general body weakness	Spray tse-tse fly with insecticide, rear resistant breed
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General Effects of Pests & Diseases

- a. Reduction in yield and productivity of crops and plants
- b. Reduction in quality of farm produces
- c. Increase in cost of production in the course of controlling them
- d. Reduction in income of farmers
- e. They render farm produce unattractive and unmarketable
- f. They cause malformation in plants and animals.
- g. They cause death of plants and animals.

General Control of Pests & Diseases

- a. **Cultural control** – use of farm practice to prevent or control pests and diseases e.g crop rotation, regular weeding, early harvesting, and use of resistant varieties.
- b. **Biological control** - The introduction of natural enemies of pests to control the pests.
- c. **Physical control** - Physical removal of pest by hand – picking, setting traps, shooting or fencing a farm with wire nets.
- d. **Chemical control** - Use of chemicals called pesticides to control pest.

